# S. Nazifa Tasnia

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## EDUCATION

**Bachelor of Engineering- Mechanical Engineering (specialization in Mechatronics)** *Graduated* May 2025
Toronto Metropolitan University (Formerly Ryerson University)

## CORE COMPETENCIES

**Technical Tools:** Python | SolidWorks | MATLAB | Allen Bradley PLC | MS Office | Alteryx | Tableau
**Manufacturing & Design:** DFMA | ASME Y14.5M |
**Quality & Process:** FMEA | ISO 55001 | OHSA Compliance

## WORK EXPERIENCE

**Planning Integration and Analytics Senior Technical Student (Co-op)** May 2023 – August 2024
Toronto Hydro Electric System Limited| Toronto, Canada

- Supported the 2025–2029 Electricity Distribution Rate Application by conducting variance analyses and aligning capital plans with regulatory requirements, contributing to successful regulatory filing.
- Reduced response time by 15% by building an Alteryx-based tracker to consolidate and manage ~1200 interrogatories.
- Delivered a GIS-based Ward Analysis for ~3,000 infrastructure projects using Alteryx, Python fuzzy matching, and Tableau, achieving a 93% match rate.
- Facilitated cross-functional collaboration with IT and consultants during ISO 55001 and Project Portfolio Management UAT initiatives.
- Authored process documentation for asset management systems, growing internal ISO 55001 certification efforts.

## ENGINEERING PROJECTS

**Robotic Manipulator for the Food Industry** April 2025
Toronto Metropolitan University | Capstone Project

- Designed a glove-controlled robotic arm with a soft gripper for automated produce handling, applying DFMA principles to optimize structure for hygiene, ease of cleaning, and assembly.
- Performed FEA in SolidWorks to assess gripper stress distribution under load; modeled gripper compression and grip force based on produce weight.
- Simulated pick-and-place sequences using Python and inverse kinematics for trajectory planning across 3 DoF.
- Implemented motion control algorithms with time-optimized path transitions for improved cycle time and grip.

**Custom Drilling Jig and Functional Gage Systems** April 2025
Toronto Metropolitan University | Course Project: Topics in Manufacturing

- Constructed a modular drill jig using the 3-2-1 locating principle to constrain cylindrical parts, supported by V-blocks and machined walls.
- Applied GD&T feature tolerances using the 50% rule and selected MIC-6 aluminum and 1018 steel for structural and thermal stability.
- Formulated a GO/NO-GO inspection gage with tolerance stack-up analysis for hole validation and feature control.
- Produced full assembly and exploded CAD drawings using SolidWorks and selected off-the-shelf components to ensure practical build feasibility.

**Motor Housing Bearing Installer** November 2024
Toronto Metropolitan University | Course Project: Manufacturing System Control

- Programmed a PLC-controlled pneumatic system using Allen-Bradley MicroLogix 1400 for automated installation of motor housing bearings using PLC logic and I/O management.
- Evaluated system risks including hazard identification, component selection, and layout planning compliant with OHSA standards.

**Inverted Pendulum Stabilization System** November 2024
Toronto Metropolitan University | Course Project: Mechatronics Systems Design

- Configured and tuned a closed-loop PID control system to stabilize a dynamic inverted pendulum.
- Modeled system dynamics in Python and tuned PID gains to achieve fast, stable responses.
- Integrated hardware and applied FMEA to validate system reliability and mitigate control risks.